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Jharkhand University of Technology, Ranchi
Diploma 2nd Semester Examination, 2025 (NEP)

Subject : Statistic and Analytics
Time Allowed : 3 Hours

Subject Code : AEC 201
Full Marks : 70
Pass Marks : 21

Answer in your own words.

Answer any five questions in which Question No. 1 is compulsory.

The figures in the margin indicate full marks.

All questions carry equal marks.

2×7=14

1. Write correct answer of the following questions:

(i) Which of the following is a data collection tool?

(a) Histogram ✓

(b) Pie Chart ✗

(c) Questionnaire ✓

(d) Box Plot ✗

(ii) Which type of data is continuous?

(a) Number of students

(b) Temperature ✓

(c) Gender ✓

(d) Blood group ✗

(iii) Which chart is used to represent relative frequency?

(a) Bar chart ✓

(b) Pie chart ✗

(c) Frequency polygon ✓

(d) Relative frequency polygon ✗

(iv) Which measure is not a measure of central tendency?

(a) Mean ✓

(b) Median ✓

(c) Mode ✓

(d) Variance ✓

(v) Which function is used to read data from a file in Python?

(a) write()

(b) read() ✓

(c) open()

(d) input()

(vi) Which of the following is a valid Python data type?

(a) integer ✓

(b) number

(c) float

(d) decimal

(vii) Which Python loop executes at least once?

(a) for

(b) while

(c) do-while

(d) Python does not have do-while ✓

Please Turn Over

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(2)

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2. (a) Define qualitative and quantitative data with examples. ✓
(b) Explain the process of data cleaning and its importance. 7+7
3. (a) Describe how to create a bar chart and pie chart using Microsoft Excel.
(b) Explain the concept of frequency polygon and frequency curve. 7+7
- ✓ 4. (a) Define mean, median and mode. Explain how to calculate them in Excel.
(b) What is standard deviation? How is it different from variance? 7+7
5. (a) Write a Python program to read a text file and count the number of lines.
(b) Explain the use of if-else and loops in Python with examples. 7+7
6. (a) What are arrays in Python? Write a program to find the sum of array elements.
(b) Define a function in Python to calculate factorial using recursion. 7+7
7. Write short notes on any four of the following 3 5/4 14
 - ✓ (a) Types of Data ✓
 - ✓ (b) Box Plot ✓
 - (c) Skewness and Kurtosis
 - (d) Python Comments ✓
 - (e) File Opening Modes in Python •
 - ✓ (f) Frequency Table